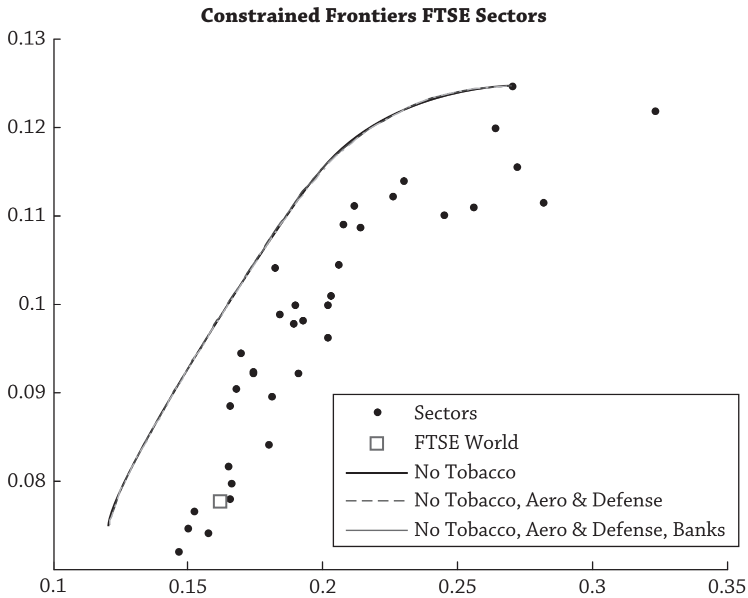

Figure 3.17

The extremely small costs of divestment are not due to the portfolios having zero holdings in the sectors that are being removed. In the full universe, the portfolio with the maximum Sharpe ratio contains 1.53% tobacco, 1.19% aerospace and defense, and 9.52% banks. Even removing an approximately 10% bank holding position has negligible cost in terms of diversification losses.

Diversification losses are so small because extra diversification benefits going from thirty-eight to thirty-nine sectors, or even thirty-six to thirty-nine sectors, are tiny (recall there are *decreasing marginal diversification benefits*). In section 2 when we added Germany and France to the G3 (United States, Japan, and United Kingdom), there was a much smaller shift in the frontier compared with moving from the United States–Japan to the G3 (see Figures 3.4 and 3.5). In our sector example the small marginal diversification benefits come about because there are few opportunities for that lost sector to pay off handsomely when the other thirty-eight sectors tank.

6.2. SOCIALLY RESPONSIBLE INVESTING

From the mean-variance investing point of view, SRI must always lose money because it reduces diversification benefits. Could it make money as an active management (alpha) strategy? Studies like Kempf and Osthoff (2007) find that stocks that rank highly on KLD measures have high returns. MSCI, an index provider, considers various social and environmental criteria in ranking companies in its